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ADVANCED PCB PROJECT

Smart Home Hub

Four-layer STM32F746 environmental, connected-weather and alert platform

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AS-BUILT TECHNICAL REPORT · REVISION 2.1

Document summary

Field	Value
Document	Smart Home Hub — Advanced PCB Project Technical Report
Revision / date	2.1 / 16 August 2026
Project stage	Manufactured, assembled and integrated prototype; full core system demonstrated
Design tools	Altium Designer; STM32CubeMX / STM32CubeIDE
Board	Four-layer rigid PCB: Top signal / GND plane / 3V3 plane / Bottom signal
Primary MCU	STM32F746BGT6

Abstract

This report documents the design, manufacture, assembly and bench verification of a smart-home hub developed for the Advanced PCB laboratory. The custom four-layer board is centred on an STM32F746BGT6 microcontroller and integrates a 4.3-inch RGB display, SCD41 and BME688 environmental sensors, a flame-detection front end, an audible alarm, Wi-Fi connectivity, a USB service interface and two external memories.

The final system presents location-aware outdoor information and the important indoor measurements on a QSPI-themed graphical dashboard. The SCD41 provides the compensated room temperature, relative humidity and CO_2 concentration; the BME688 provides pressure and relative gas-resistance diagnostics. Flame and persistent gas-change monitoring remain active in the background and can raise a priority on-screen warning and sound the buzzer independently of the cloud connection.

Bench testing confirms the protected 5 V input, regulated 3.3 V rail, MCU, RGB display and backlight, SCD41, BME688, flame detector, buzzer, ESP-01 Wi-Fi/Blynk link, internet time/date/outdoor temperature, W25Q128 QSPI flash, the full 8 MB IS42S16400 SDRAM and the MCP2221A USB-to-UART console. The USB bridge originally failed because the USBLC6-2SC6 D+ protection path was shorted to ground; after rework it enumerated and carried bidirectional USART3 traffic. The only remaining unavailable function is touch input because the display assembly's external GT911 flex tail is mechanically torn.

SAFETY SCOPE
This is an educational prototype. The flame and gas-change functions are not certified fire, gas-leak or life-safety detectors and must not replace approved alarms.

Project outcome at a glance

Area	Status	Evidence
Core environmental dashboard	**PASS**	Live compensated SCD41 values and BME688 pressure shown on the RGB display
Local alarm path	**PASS**	Priority warning screen and controlled buzzer verified
Cloud and weather data	**PASS**	Blynk plus Brussels outdoor temperature, time and date
External graphics storage	**PASS**	W25Q128 ID EF 40 18; themed backgrounds, fonts and icons
External frame memory	**PASS**	All 8 MB repeatedly verified and used for GUI double buffering
USB service console	**PASS**	MCP2221A enumerates and exchanges terminal commands over USART3
Capacitive touch	**NOT TESTABLE**	External GT911 FPC mechanically torn

Figure 1 — Final working dashboard with network/cloud status, outdoor temperature, indoor measurements and evaluated air quality.

Contents

READING GUIDE
Sections 4–7 explain the implementation. Sections 8–10 contain the strongest evaluation evidence because they distinguish verified behaviour from remaining technical risks.

1 Introduction and scope

A single PCB that senses, informs, communicates and warns

The Smart Home Hub was conceived as a compact wall-mounted room controller and environmental monitor. It collects indoor measurements, presents them locally on a 480 × 272 display, publishes selected values remotely, retrieves location-aware outdoor information and raises an alert when an abnormal condition persists. The project deliberately combines digital buses, analog sensing, a high-pin-count RGB interface, switch-mode power conversion, external memory, USB and radio communication so that the PCB exercises the complete Advanced PCB workflow.

1.1 Functional scope

- Measure CO_2 concentration, temperature and humidity using the SCD41 on I^2C4.
- Measure barometric pressure and relative gas resistance using the BME688 on I^2C3; retain its temperature as a PCB/heater diagnostic.
- Monitor a phototransistor flame channel through ADC1/PA1 and an active-low comparator output on PH2.
- Show compensated room temperature/humidity, CO_2, pressure, air-quality interpretation, Wi-Fi/cloud state, local time/date and outdoor temperature on the 4.3-inch RGB display.
- Maintain flame and persistent gas-change monitoring in the background; display a warning and sound the buzzer when alarm criteria are met.
- Connect the ESP-01 through USART1 for Wi-Fi, Blynk and weather/time HTTP; use the MCP2221A on USART3 for PC service commands.

1.2 Deliberate limitations

The board is a laboratory prototype rather than a certified appliance. No claim is made for calibrated gas identification, fire-alarm compliance, long-term environmental accuracy, EMC or electrical-safety certification. The BME688 gas channel is a relative baseline-change detector and does not report a validated gas concentration in ppm. Plain HTTP is used only as a laboratory compatibility workaround for the installed ESP-01 firmware.

1.3 Evaluation evidence

Evidence comes from the archived Altium project, production data, schematic and PCB documentation, assembly observations, firmware source and repeatable bench tests. PASS is reserved for an observed end-to-end path. The GT911 touch function is marked NOT TESTABLE because its external flex is torn rather than because the PCB interface failed.

2 Requirements and success criteria

ID	Requirement	Acceptance evidence	Status
R1	5 V input and stable 3.3 V rail	Power-up without overheating; regulated rail present	PASS
R2	STM32 boots and can be programmed	SWD programming and integrated firmware execution	PASS
R3	Environmental sensing	Credible SCD41 and BME688 data	PASS
R4	Local display	Legible dashboard; smooth double-buffered updates	PASS
R5	Abnormal-condition alert	Priority screen and controlled buzzer	PASS
R6	Wi-Fi/cloud link	ESP-01 joins AP; Blynk update succeeds	PASS
R7	QSPI asset storage	JEDEC ID plus read/write and asset loading	PASS
R8	SDRAM	Repeated full 8 MB test at 0xC0000000	PASS
R9	Capacitive touch	GT911 enumeration and coordinate reports	NOT TESTABLE
R10	USB-to-UART contingency	COM enumeration and bidirectional terminal commands	PASS

DEFINITION OF DONE
The prototype is demonstrable when the MCU, display, environmental sensors, local alarm, Wi-Fi telemetry, memories and PC service interface operate together. Touch remains an external mechanical limitation and is disclosed separately.

3 System architecture

Domain separation simplifies design, bring-up and fault isolation

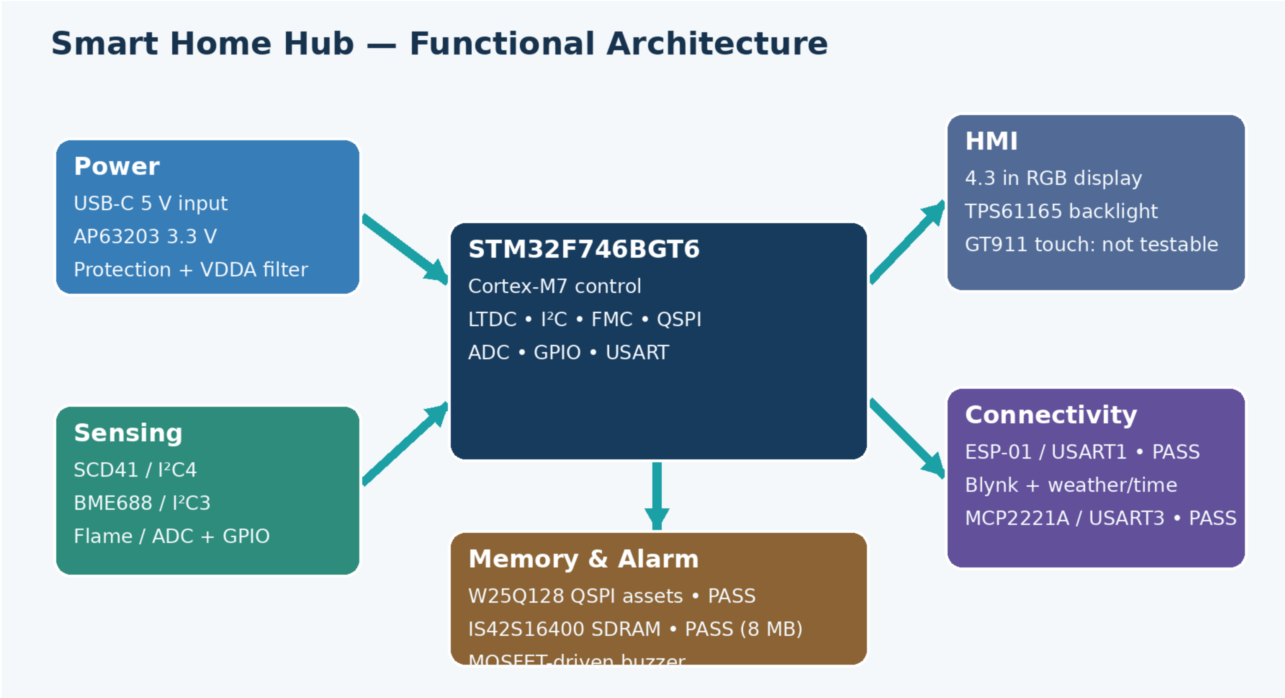


Figure 2 — Functional architecture and final verification state.

3.1 Interface allocation

MCU resource	Connected block	Purpose
LTDC / RGB	MDT0430A01ISC-RGB	480 × 272 parallel pixel data and timing
I²C4	SCD41	Compensated room CO_2, temperature and humidity
I²C3	BME688	Pressure, gas resistance and PCB thermal diagnostics
ADC1 IN1 / PA1	Flame A0	Continuous analog optical response
GPIO PH2	Flame D0	Active-low comparator alarm
USART1	ESP-01	AT commands, Wi-Fi, Blynk and weather/time HTTP
USART3	MCP2221A	Bidirectional USB PC service console
QUADSPI	W25Q128JV	Background themes, font glyphs and icons
FMC SDRAM	IS42S16400J	8 MB framebuffers and working memory
GPIO	Buzzer MOSFET	Default-off low-side alarm switching

4 Hardware design

Power, processing, sensing, HMI, alarm, connectivity and memory

4.1 Power entry, protection and conversion

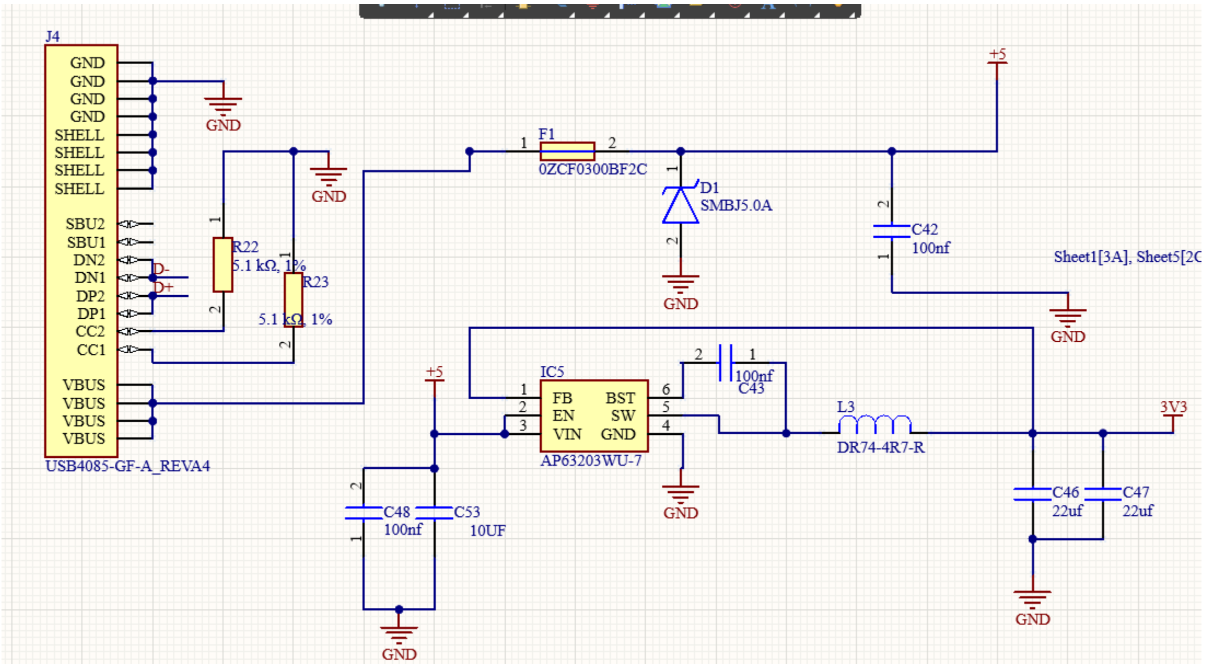


Figure 3 — USB-C power entry, protection and AP63203 3.3 V buck converter.

The hub is powered from a 5 V adapter rated for 3–5 A. The USB-C receptacle is configured as a sink with 5.1 kΩ pull-downs on CC1 and CC2. Joined VBUS pins feed a resettable fuse and 5 V TVS device before the protected rail reaches the load. The AP63203 synchronous buck converts 5 V to 3.3 V using local ceramic input, bootstrap and output capacitors and a shielded inductor.

The 5 V rail also supplies the display backlight and buzzer. The 3.3 V plane supplies the STM32 digital domain, sensors, memories, MCP2221A I/O and ESP-01. A ferrite-bead branch supplies VDDA to reduce switching noise before the ADC/reference domain.

> **BRING-UP RULE**
> Fit the MCU decoupling capacitors before functional testing. A Cortex-M7 may appear to boot without every 100 nF capacitor, but that is not a robust or valid hardware test.

4.2 MCU supply, clocking and display backlight

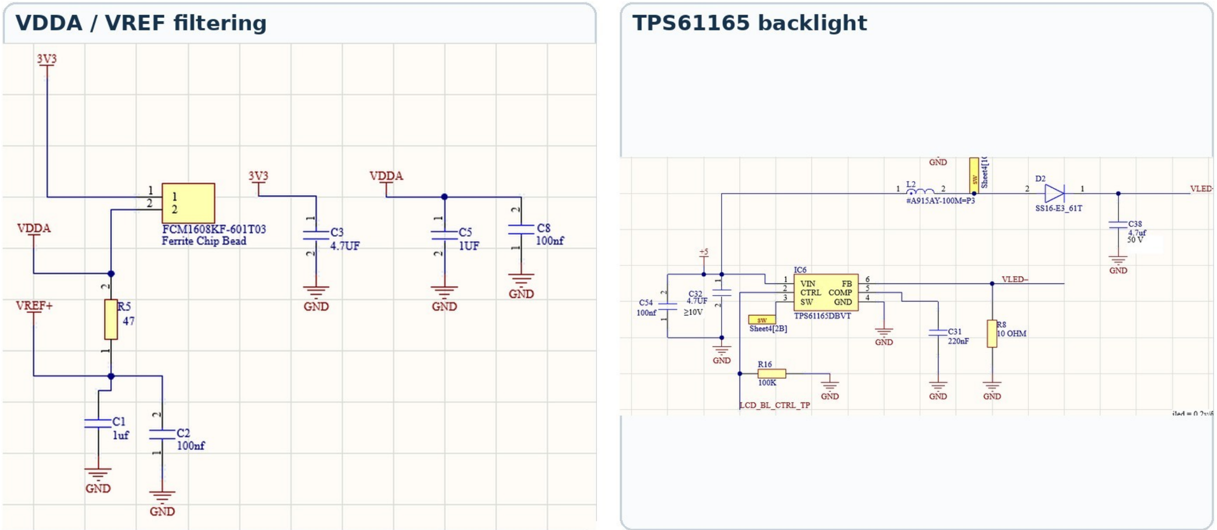


Figure 4 — Analog-supply filtering and the TPS61165 constant-current backlight stage.

MCU and analog reference

The STM32F746BGT6 is powered at 3.3 V with one 100 nF ceramic capacitor per VDD group and nearby bulk capacitance. VDDA is fed through the FCM1608KF-601T03 ferrite bead and locally decoupled. VREF+ is derived through 47 Ω and separately filtered, providing a quiet reference for the flame-sensor ADC.

The clock tree uses a 32.768 kHz low-speed crystal and an 8 MHz oscillator module. The final system runs at 200 MHz HCLK, while the FMC SDRAM clock is approximately 66 MHz.

Display and backlight

The 40-contact display flex carries 24-bit RGB, pixel clock, display enable and synchronization. The TPS61165 regulates LED-string current. CTRL is driven by the STM32 and has a 100 kΩ pull-down, so the backlight remains off while the MCU is unpowered. The boost output capacitor is rated for the LED-string voltage.

ASSEMBLY CORRECTION
The first backlight test failed because the external Schottky diode was fitted with reversed polarity. Correcting its orientation restored operation; the 4.7 µF / 50 V output capacitor was not the root cause.

4.3 Environmental sensing and compensation

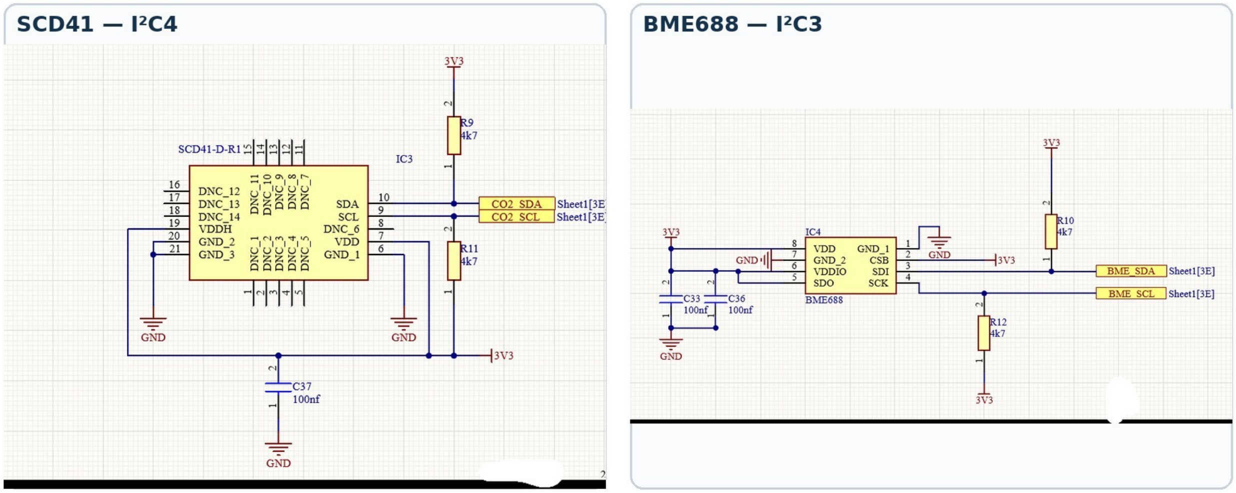


Figure 5 — Separate I²C buses and local pull-ups for the SCD41 and BME688.

The SCD41 is powered from 3.3 V and connected to I²C4 with 4.7 kΩ pull-ups. It is the authoritative source for room CO_2, temperature and relative humidity. Because the assembled display/PCB heats the local air, firmware programs a board-specific 9.3 °C SCD4x temperature offset at boot and applies 1/4 IIR smoothing. The compensated SCD41 values drive the dashboard, Blynk V4/V5 and room temperature/humidity alarms.

The BME688 is connected to I²C3 with independent VDD/VDDIO decoupling. CSB selects I²C mode, and firmware checks addresses 0x76 and 0x77. Pressure and gas resistance are used operationally. BME688 temperature remains visible in diagnostics as an indicator of PCB/heater conditions rather than the displayed room temperature.

Relative gas-change interpretation

After 60 valid heater-stable samples at 5 s intervals (about 5 min in clean air), the firmware establishes a baseline. An alarm requires a persistent change of roughly 30% from baseline for four consecutive samples (about 20 s). It clears only after the value returns within roughly 15% for six samples (about 30 s). This persistence and hysteresis suppress nuisance alarms while retaining sensitivity to a sustained change.

CALIBRATION CAUTION
The 9.3 °C SCD41 offset is installation-specific and must be recalibrated if the enclosure, display heating, airflow or sensor placement changes. Neither sensor result is a traceable calibration certificate.

4.4 Flame detector and audible alarm

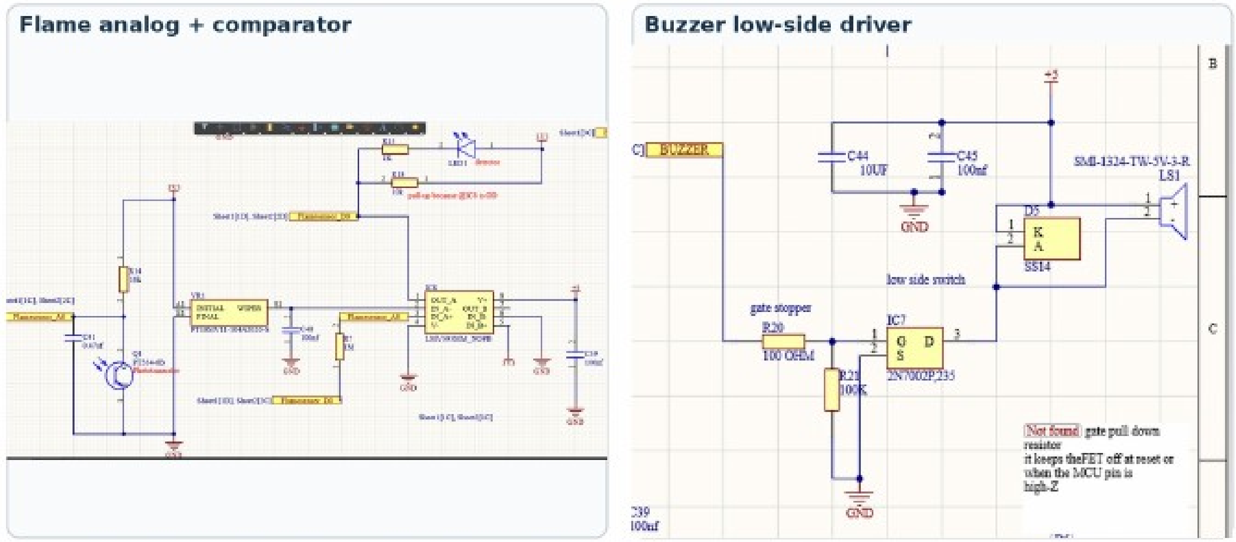


Figure 6 — Dual analog/digital flame path and default-off buzzer driver.

The PT334-6B phototransistor creates a light-dependent voltage at Flamesensor_A0. A 10 kΩ pull-up converts photocurrent to voltage and a 0.47 µF capacitor reduces high-frequency noise. PA1 samples this node using ADC1. The LMV393 compares the signal with the potentiometer threshold and generates an open-drain active-low D0 signal on PH2; a 10 kΩ pull-up defines the logic level and a local LED indicates detection.

The buzzer is supplied from 5 V and switched on the low side by a 2N7002 MOSFET. A 100 Ω gate resistor limits edge current, while the essential 100 kΩ gate-to-ground resistor keeps the MOSFET off during reset so the buzzer does not sound immediately at power-up.

Condition	Analog observation	Comparator	System response
No flame	A0 ≈ 3.19 V / ADC ≈3950	D0 high	Buzzer off
Lighter presented	A0 ≈ 1.5 V / ADC ≈1730	D0 low	Priority alarm logic active

SENSOR LIMITATION
A phototransistor can respond to sunlight, lamps and infrared sources. The dual analog/digital path aids diagnosis but is not a certified flame detector.

4.5 Connectivity and service interfaces

Interface	Electrical link	Purpose	Observed state
ESP-01	USART1	Wi-Fi AT command link	PASS — WLAN, TCP, Blynk and weather/time
MCP2221A	USB D+/D−; USART3	PC service/debug console	PASS — enumerates and exchanges commands
SWD	SWDIO / SWCLK / NRST	Programming and debug	PASS — essential bring-up path
Test pads	Power/control nets	Probe access and rework	Included where practical

ESP-01 and cloud path

The ESP-01 operates at 3.3 V on USART1. Firmware confirms AT operation, joins the configured access point, opens TCP connections, updates Blynk and retrieves Brussels weather/time data. The working laboratory build uses plain HTTP on port 80 because the installed ESP AT firmware could not complete current TLS negotiation.

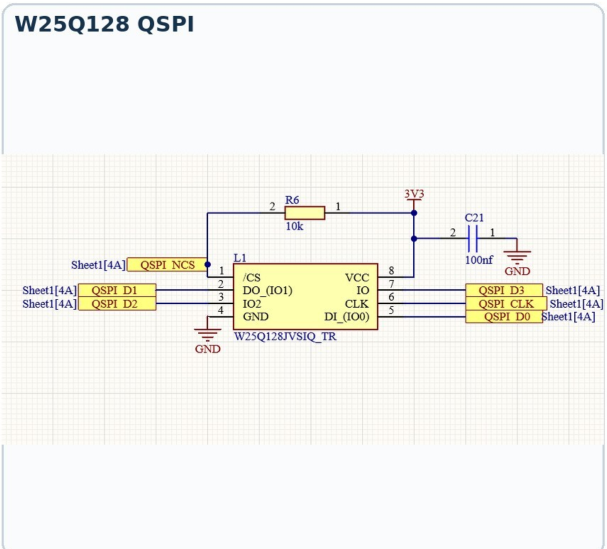
MCP2221A USB service console — resolved

The MCP2221A initially stopped enumerating and D+ measured shorted to ground. Continuity inspection isolated the fault to the USBLC6-2SC6 protection pin/path. After removing the short/reworking the device, Windows enumeration returned, the Microchip utility read the bridge configuration and bidirectional UART traffic was verified. PC10/USART3_TX drives MCP2221A URX; PC11/USART3_RX receives MCP2221A UTX at 115200 8N1.

The service console accepts HELP, PING, INFO, STATUS, SENSORS, NETWORK and MEMORY. A PING→PONG round trip proves the complete PC→USB→MCP2221A→STM32→MCP2221A→USB→PC path.

SECURITY IMPROVEMENT
Plain HTTP is acceptable only as a laboratory diagnostic. A deployable product should use current TLS, certificate validation and secure credential provisioning.

4.6 External memory



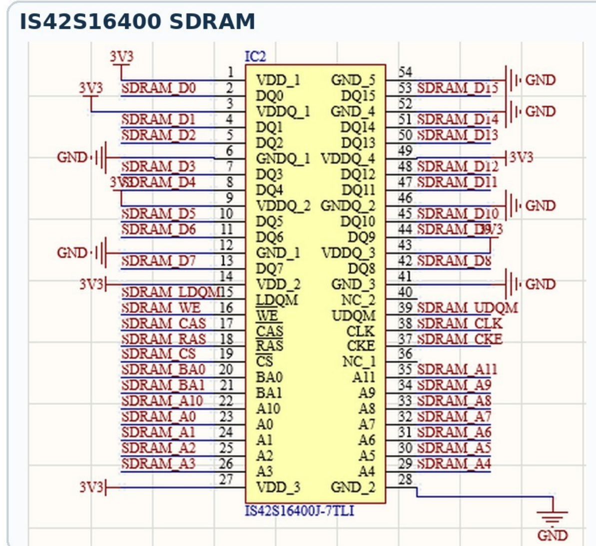


Figure 7 — External W25Q128 QSPI flash and 16-bit IS42S16400 SDRAM interfaces.

W25Q128JV QSPI flash

The 128-Mbit W25Q128JV provides 16 MB of non-volatile storage. /CS has a 10 kΩ pull-up so the flash remains deselected during reset. Firmware read JEDEC ID EF 40 18 and verified erase/program/read in a reserved region. The final build stores time/condition-themed backgrounds, the modern font and GUI icons in QSPI.

IS42S16400J SDRAM

The 64-Mbit SDR SDRAM is connected to the FMC through a 16-bit data bus plus address, bank, byte-mask, clock, enable and command signals. Startup issues clock-enable, precharge-all, auto-refresh and mode-register commands before programming refresh. Rework and a clean CubeMX FMC pin/timing configuration converted an early mismatch into a repeatable full-capacity PASS.

Item	Verified result
Base address	0xC0000000
Capacity exercised	8 MB / complete address range
Diagnostic clock	Approximately 66 MHz
Test	Address + multiple data patterns, repeated
Production use	Two 480 × 272 L8 GUI buffers plus working memory

MEMORY ROLE SEPARATION
QSPI is persistent asset storage; SDRAM is volatile framebuffer/working memory; internal SRAM holds stacks, drivers and compact control state.

5 PCB implementation

Four-layer stack, plane strategy, routing and design-rule control

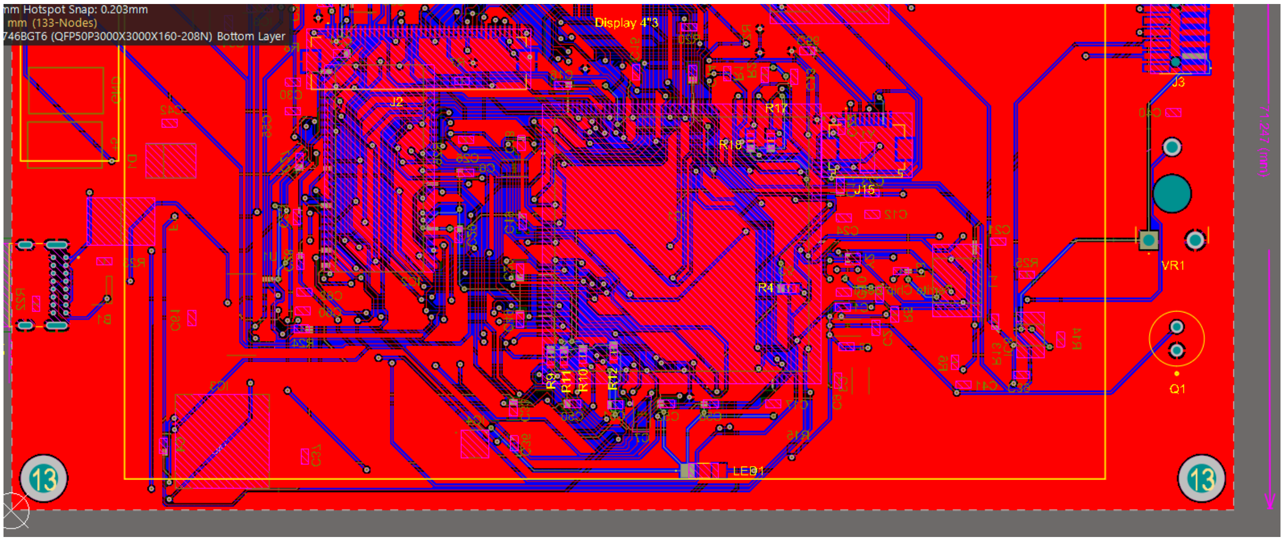


Figure 8 — Altium PCB view with top/bottom routing and internal planes visible.

Layer	Assignment	Design intent
Top	Signal + components	Dense STM32, memory and display escape routing
Layer 1	GND plane	Low-impedance return path and reference plane
Layer 2	3V3 plane	Main low-voltage distribution
Bottom	Signal + components	Secondary routing, connectors and peripheral circuits

5.1 Placement and routing strategy

The STM32, SDRAM and QSPI flash occupy the dense central region to minimize parallel-bus length. The display connector sits at the edge to suit the flex direction. Power conversion remains near USB-C, while the optical flame sensor and threshold control are near the perimeter. Situs autorouting provided a starting point, followed by visual review and manual cleanup of power distribution, plane connections and constrained areas.

- Reference high-frequency signals to an uninterrupted ground plane and avoid return-current detours.
- Place each 100 nF decoupler beside its supply pin group with a short pad-to-via loop.
- Keep the backlight SW/inductor loop compact and away from analog/sensor signals.
- Use wide copper for 5 V backlight/buzzer current and the 3.3 V regulator output.

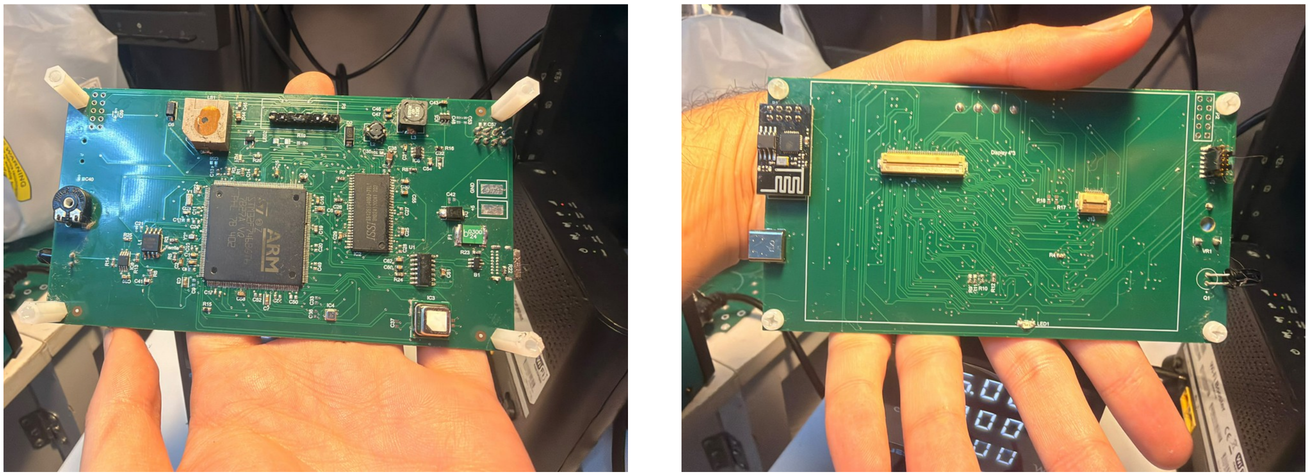
5.2 Design-rule and manufacturing checks

The board was checked for electrical clearance, routing width, via geometry, solder-mask slivers, silkscreen-to-mask clearance, component clearance and power-plane relief. Rules that generated excessive warnings were aligned with the selected fabricator's published capability; individual geometry errors were corrected rather than hidden by globally disabling checks.

Review item	Final treatment
Copper clearance / width	Checked against the selected fabricator's limits
Solder-mask slivers	Rule aligned with process capability; dense-pad cases manually reviewed
Silkscreen clearance	Overlay moved where it could interfere with soldering
Plane thermal relief	Blocked-spoke warnings reviewed at USB-C and plane-connected pads
Hole geometry	Four overlapping NPTH features corrected after CAM feedback

PLANE CONNECTIVITY
Layer 1 is assigned to GND and Layer 2 to 3V3. Plane-net assignment, repour, zero unrouted nets and final DRC are all required; a visually filled plane alone does not prove connectivity.

6 Manufacture and assembly



(a) Component side of the functional PCB.

(b) Rear/connector side with display interfaces, ESP-01 and USB.

Figure 9 — Assembled four-layer prototype, shown from both sides.

6.1 Production package

Output	Purpose
Gerber / layer outputs	Copper, mask, overlay, paste and mechanical data
NC drill	Plated, non-plated, round and slot holes
Pick-and-place + BOM	Centroids/orientations and fitted-part information
Draftsman	Fabrication, assembly and drill drawings
DRC / ERC / differences	Design verification records
STEP	Board-level 3D exchange model

6.2 Fabrication query and assembly lessons

The fabricator identified four overlapping non-plated holes and asked whether the final diameter should be 3.45 mm or 3.2 mm. The source geometry was corrected before production. The most important component-level faults were also diagnosed systematically: the reversed TPS61165 Schottky diode, FMC/SDRAM solder/configuration issues and the USBLC6-2SC6 D+ short.

7 Firmware and graphical interface

Non-blocking acquisition, compensated values, local alarms and external-memory graphics

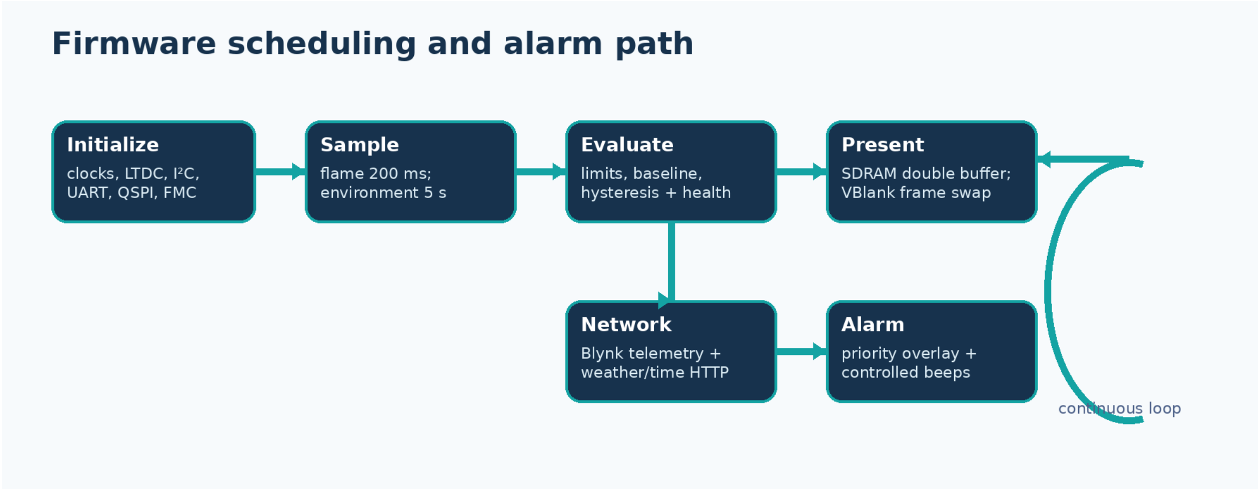


Figure 10 — Cooperative time-based scheduler used by the integrated firmware.

7.1 Peripheral configuration

Peripheral	Role
LTDC	480 × 272 RGB timing with SDRAM-backed L8 layers
I^2C3 / I^2C4	BME688 and SCD41 on separate buses
ADC1 / GPIO	Flame analog PA1, D0 PH2 and buzzer control
USART1	ESP-01 AT commands and internet services
USART3	MCP2221A bidirectional PC service console
QUADSPI	W25Q128 background/font/icon assets
FMC	8 MB SDRAM diagnostics and GUI framebuffers

7.2 Timing and fault isolation

HAL_GetTick() schedules the cooperative main loop. Flame sampling runs every 200 ms; environmental acquisition approximately every 5 s; GUI refresh every 500 ms; Blynk upload every 10 s; Wi-Fi retry every 30 s; and weather/time retrieval at a slower interval. Each service owns an independent status so a failed cloud request does not disable local sensing or alarms.

7.3 Final graphical dashboard

Figure 11 — Final QSPI-themed, SDRAM-double-buffered dashboard with live local and internet data.

The top bar shows the clock, Wi-Fi/cloud icons and date. The centre shows the configured location and outdoor temperature. Indoor values use a two-column layout: compensated SCD41 room temperature/humidity on the left, and SCD41 CO_2 plus BME688 pressure on the right. The air-quality state and explanation remain visible below. Raw flame, gas and BME temperature continue in diagnostics/background monitoring and do not clutter the normal screen.

Flicker was removed by drawing the next dynamic layer into the inactive 480 × 272 L8 buffer in SDRAM, cleaning the relevant Cortex-M7 cache range and swapping the LTDC layer address during vertical blanking. QSPI supplies the themed background, modern font and icons; only a complete, ready frame is presented.

MEMORY USAGE
Two L8 GUI buffers consume about 255 KiB of SDRAM. QSPI supplies persistent assets without enlarging internal flash, and internal SRAM remains available for the scheduler, network parser and drivers.

7.4 Cloud data and alarm policy

Blynk datastream	Content	Representation
V0–V2	Flame state, ADC and voltage	Boolean / raw / V
V3	SCD41 CO_2	ppm
V4 / V5	Compensated SCD41 room T / RH	°C / %RH
V6 / V7	BME688 temperature / humidity diagnostics	°C / %RH
V8	BME688 pressure	hPa
V9–V11	Gas resistance, ratio and alarm	Ω / % / Boolean

Measurements are batched where possible; flame state changes can be sent immediately. A separate HTTP request obtains Brussels outdoor temperature and the HTTP Date header supplies local date/time. The parser anchors on the actual HTTP/1.x header within the ESP-01 transcript, skips the header terminator and parses the signed body value.

Alarm philosophy

- Local indication and buzzer control never depend on Wi-Fi or Blynk availability.
- SCD41 compensated values drive room temperature/humidity decisions; BME temperature is diagnostic only.
- Gas detection arms only after a stable baseline and uses persistence plus clear hysteresis.
- The physical MOSFET gate pull-down guarantees buzzer-off during reset.
- Flame has priority; severe persistent environmental conditions can also produce a priority screen and buzzer pattern.

8 Verification and test results

Bench evidence for every major subsystem

8.1 Incremental method

Power rails and control pins were measured before firmware integration. Each digital peripheral was first tested with a narrow diagnostic program and only then added to the combined dashboard. The display was an immediate local test surface; the MCP2221A/USART3 console provided persistent diagnostic responses.

- Power — verify 5 V, regulated 3.3 V, acceptable current and repeatable MCU boot.
- Sensors — confirm I^2C acknowledgement, manufacturer command sequence and plausible live values.
- Flame/buzzer — compare ADC voltage and D0/LED state, then verify controlled sounder behaviour.
- Connectivity — verify ESP AT responses, WLAN, Blynk, weather/time parsing and MCP terminal round trips.
- Memory — read QSPI ID and repeat complete 8 MB SDRAM address/data-pattern tests.
- HMI — verify backlight, QSPI assets, cache maintenance, VBlank buffer swapping and alarm priority.

8.2 Integrated demonstration snapshots

Snapshot	Outdoor	Room T / RH	CO_2	Pressure	Air quality
Normal GUI photo	22 °C	34.3 °C / 31.9 %	675 ppm	1005.5 hPa	GOOD
Second stable state	24 °C	26.3 °C / 52.7 %	600 ppm	1009.2 hPa	GOOD

INTERPRETATION
These snapshots prove the complete acquisition/display path. They are not calibration certificates. The installation-specific SCD41 compensation should be verified against a reference after enclosure thermal equilibrium.

8.3 Verification matrix

Subsystem	Status	Evidence
5 V input / AP63203 3.3 V	PASS	Rails present; MCU/peripherals operate
STM32F746 boot / SWD	PASS	Firmware programs and runs
RGB LCD + TPS61165	PASS	Image and backlight stable after diode correction
SCD41 on I^2C4	PASS	Compensated CO_2/T/RH integrated
BME688 on I^2C3	PASS	Pressure/gas and diagnostics acquired
Flame A0 / D0	PASS	ADC and active-low D0 respond
Buzzer driver	PASS	Controlled by STM32; silent at reset
ESP-01 / Blynk / weather	PASS	AT, AP, TCP, Blynk and outdoor data
W25Q128 QSPI	PASS	EF 40 18; verified data and assets
IS42S16400 SDRAM	PASS	All 8 MB repeatedly verified at ~66 MHz
MCP2221A USB-UART	PASS	USBLC6 fault repaired; enumeration and two-way commands
GT911 touch	NOT TESTABLE	External touch FPC completely torn

8.4 Recommended demonstration sequence

1. Show the stable dashboard and explain the compensated indoor values.
2. Present a controlled flame or temperature alarm and the buzzer response.
3. Show a Blynk update and a weather/time refresh through the ESP-01.
4. Open the MCP2221A COM port and demonstrate PING→PONG plus SENSORS or MEMORY.
5. Run QSPI identification and the repeated full 8 MB SDRAM diagnostic.
6. Show the archived Altium source, layer stack, DRC/ERC, Draftsman, BOM, STEP and factory outputs.

8.5 Priority alarm evidence

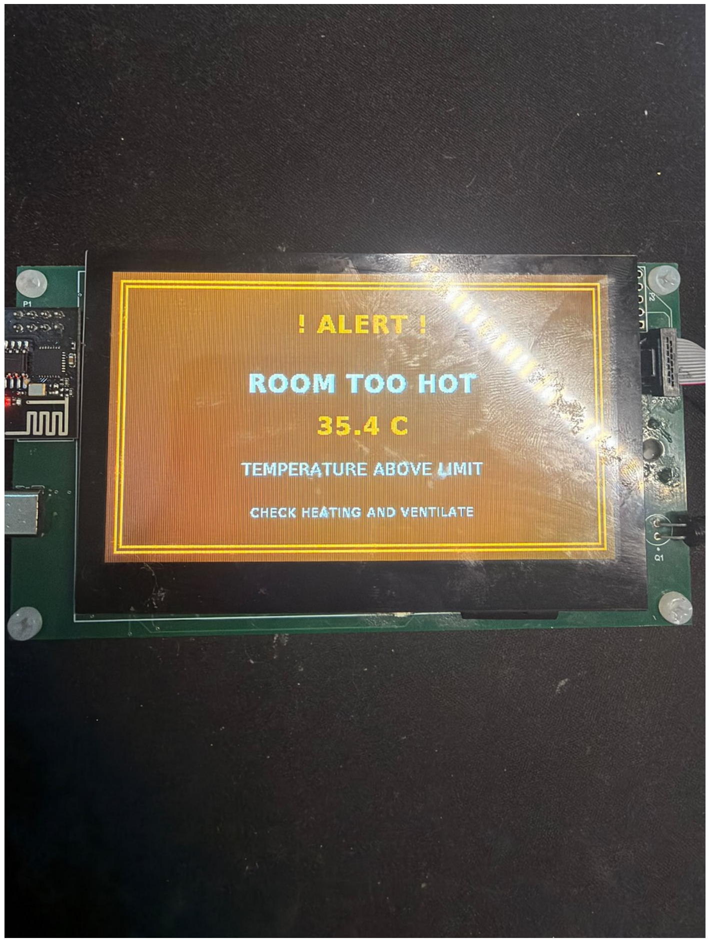


Figure 12 — Persistent high-temperature condition produces a priority warning screen and simultaneously activates the controlled buzzer.

The screen explicitly identifies the active condition, measured value and recommended response. The dashboard is restored after the alarm clears. This behaviour verifies the complete sensor decision → GUI priority → GPIO/MOSFET → buzzer chain rather than only an isolated buzzer test.

SAFETY STATEMENT
Alarm functions are valuable course demonstrations but remain non-certified. The hub must not replace approved smoke, flame, gas or carbon-monoxide alarms.

9 Problems, corrections and lessons

Failures were retained as structured engineering evidence

Symptom	Cause / status	Action	Rev-B prevention
Backlight remained off	External Schottky reversed	Corrected polarity	Larger cathode marking + checklist
Display flashed	Whole frame exposed while drawing	SDRAM double buffer + VBlank swap	Retain cache/VBlank discipline
ESP weather/SSL failed	Old TLS path; parser start wrong	HTTP and bounded HTTP-body parser	Current TLS-capable radio/firmware
MCP2221A did not enumerate	USBLC6-2SC6 D+ path shorted to GND	Continuity diagnosis + rework; full UART PASS	Verify ESD footprint; isolation/loopback pads
SDRAM mismatch	FMC conflicts plus solder/contact	Rework + clean FMC config; 8 MB PASS	Preserve production self-test
Touch unavailable	External flex mechanically torn	Removed from demonstration	Replaceable FFC + strain relief
Fabricator hole query	Four overlapping NPTH features	Corrected before production	Explicit hole-to-hole review

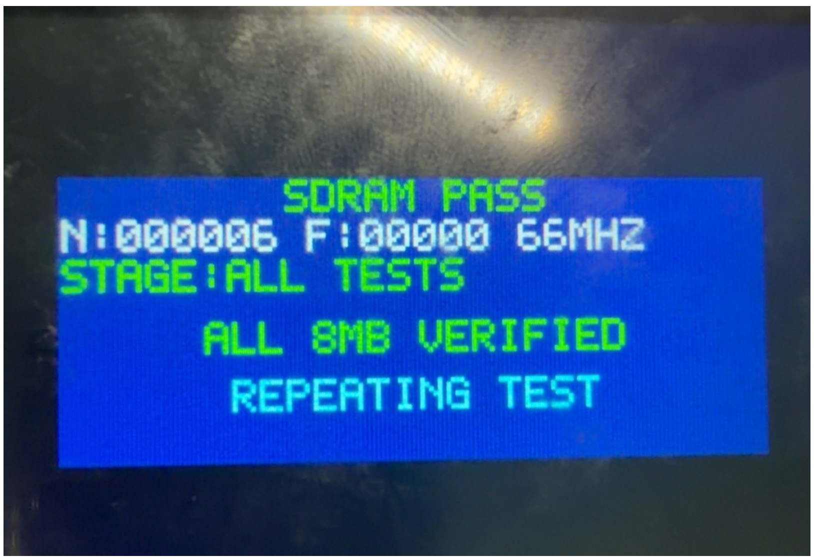


Figure 13 — Repeated full 8 MB SDRAM diagnostic after FMC correction and rework.

ENGINEERING LESSON
The SDRAM fault was not solved by adding bus pull-ups, and the MCP fault was not software. Repeatable electrical measurements, narrow subsystem tests and controlled rework converted both faults into repeatable PASS results.

10 Improvement plan for Rev B

Priority	Change	Implementation	Benefit
P1	Touch serviceability	Replaceable FFC/assembly, strain relief and no copper at bend	Restores and protects input
P1	Secure radio	Current TLS, certificate validation and supply margin	Removes plain-HTTP dependency
P2	USB robustness	Keep ESD, add labelled D+/D−/TX/RX loopback and 0 Ω isolation	Faster deterministic diagnosis
P2	Sensor thermal layout	Vented edge/peninsula away from display, MCU and inductors	Improves accuracy/response
P2	Memory robustness	Preserve 8 MB self-test and optional series footprints	Protects framebuffer path
P2	Manufacturing clarity	Stronger polarity/pin-1/NPTH checks and assembly drawings	Reduces rework/CAM queries
P3	Firmware resilience	Watchdog, fault log, alarm latching, OTA and self-test	Improves maintainability

10.1 Validation additions

A second revision should include a written test specification covering supply current/ripple, sensor warm-up, reference-instrument comparison, gas-baseline repeatability, alarm thresholds, Wi-Fi reconnect behaviour, extended run time, temperature exposure and ESD/EMI pre-compliance. The present board proves the architecture but does not provide statistical product-validation evidence.

HIGHEST-VALUE NEXT ACTION
Replace the damaged touch assembly and improve touch-flex strain relief, then repeat sensor compensation inside the intended enclosure. Preserve the passing SDRAM and MCP service tests as production self-tests.

11 Conclusion

The Smart Home Hub achieved the central objective of the Advanced PCB course: a custom manufactured four-layer PCB that boots, senses the room, presents information locally, warns through an independently controlled buzzer and communicates over both Wi-Fi and USB service links. The design integrates a high-pin-count STM32, parallel RGB display, multiple I²C buses, analog comparator/ADC path, switch-mode conversion, external serial flash, parallel SDRAM and a complete production-data workflow.

Every populated/core PCB subsystem now passes end-to-end testing: stable power, STM32/SWD, display/backlight, SCD41, BME688, flame detector, buzzer, ESP-01/Blynk, internet weather/time, W25Q128 QSPI, full 8 MB SDRAM and the repaired MCP2221A/USART3 service console. The final firmware assigns memory correctly: QSPI holds persistent backgrounds, fonts and icons; SDRAM provides double-buffered GUI layers and working space; internal SRAM retains control state.

The only unavailable feature is GT911 touch input because the display assembly's external flex tail is mechanically torn. The operating hub does not depend on that input. Plain HTTP, installation-specific temperature compensation and non-certified flame/gas algorithms remain explicit development limitations rather than hidden product claims.

FINAL ASSESSMENT
The prototype is demonstrable and functionally useful for the course. Core paths, external memories, cloud data, alarm behaviour and the USB service console are verified; the remaining touch limitation and Rev-B risks are clearly documented.

Submission talking points

- Explain the four-layer GND/3V3 plane stack and dense FMC/LTDC routing.
- Demonstrate the compensated dashboard, alarm screen and controlled buzzer.
- Show the separate QSPI asset-store and SDRAM framebuffer roles.
- Demonstrate the MCP2221A console and explain how the USBLC6 short was diagnosed.
- State the torn touch flex and safety/calibration scope without overstating the result.

Appendix A Delivered project data

ID	Deliverable	Content	State
A1	Archived Altium project	PrjPcb, SchDoc, PcbDoc, OutJob and STEP	READY
A2	Production package	Factory Gerber/layer files and NC drill	READY
A3	Smart schematic PDF	Compiled hierarchical schematic set	READY
A4	Draftsman	Assembly, fabrication and drill drawings	READY
A5	Bill of materials	Standalone XLSX export	READY
A6	Verification	DRC, ERC and differences reports	READY
A7	Firmware	STM32CubeIDE project/source and build notes	READY
A8	Technical report	This Rev 2.1 DOCX and PDF	READY
A9	Presentation	Project PPT for oral demonstration	READY
A10	A4 poster	Concise project overview	PENDING

Recommended submission folder

Folder	Contents
01_Altium_Source	Archived source project and libraries
02_Production_Data	Exact factory-accepted package
03_Schematic_PDF	Smart PDF
04_Draftsman	Assembly/fabrication/drill PDF
05_BOM	XLSX and optional PDF
06_Verification	DRC, ERC and differences
07_3D	Board STEP
08_Firmware	STM32CubeIDE project/source and notes
09_Report	DOCX and PDF
10_Presentation_Poster	PPTX and A4 poster

REPRODUCIBILITY
Keep the production package identical to the fabricator-accepted files. If outputs are regenerated later, archive them as a separate revision with an explicit reason.

Appendix B References

[1] STMicroelectronics, STM32F746BG product page and device documentation. https://www.st.com/en/microcontrollers-microprocessors/stm32f746bg.html

[2] Sensirion, SCD41 product page, SCD4x data sheet and temperature-offset guidance. https://sensirion.com/products/catalog/SCD41

[3] Bosch Sensortec, BME688 gas sensor product documentation. https://www.bosch-sensortec.com/en/products/environmental-sensors/gas-sensors/bme688/

[4] Texas Instruments, TPS61165 high-brightness LED driver data sheet. https://www.ti.com/product/TPS61165

[5] Diodes Incorporated, AP63203 synchronous buck converter. https://www.diodes.com/part/view/AP63203

[6] Microchip Technology, MCP2221A USB-to-I²C/UART bridge. https://www.microchip.com/en-us/product/mcp2221a

[7] Integrated Silicon Solution Inc., IS42S16400J SDR SDRAM family. https://www.issi.com/US/product-dram-SDR.shtml

[8] Winbond Electronics, W25Q128JV QSPI NOR flash family. https://www.winbond.com/hq/product/code-storage-flash/qspi-nor/

[9] Blynk documentation. https://docs.blynk.io/

[10] Altium Designer documentation. https://www.altium.com/documentation/altium-designer

[11] Project evidence: archived Altium design, factory outputs, schematics, STM32CubeIDE firmware, final dashboard photographs and bench observations.

Component naming note

Part numbers in this report follow the design and assembled prototype evidence available during preparation. The fabrication BOM, schematic libraries and fitted-component inspection remain authoritative for package, tolerance and ordering-suffix details.